

Commerce XI Maths App

Theory (Pages 49–52)

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 4
Bivariate Frequency Distribution and Chi Square Statistic
Total questions: 40

Question 1 - MCQ · PG-49 · TH-Q1

A variable taking non-numerical values is called

- A. categorical variable
- B. continuous variable only
- C. class interval
- D. frequency total

Correct answer: categorical variable

Explanation and steps

Categories such as gender and blood group are non-numerical.

Question 2 - MCQ · PG-49 · TH-Q2

Which is a categorical variable?

- A. blood group
- B. height in cm
- C. weight in kg
- D. income in rupees

Correct answer: blood group

Explanation and steps

Blood group consists of named categories.

Question 3 - MCQ · PG-49 · TH-Q3

A two-way frequency table for categorical variables is called

- A. contingency table
- B. ogive
- C. histogram
- D. univariate array

Correct answer: contingency table

Explanation and steps

It summarizes two categorical variables.

Question 4 - MCQ · PG-49 · TH-Q4

A contingency table is used to study

- A. two categorical variables
- B. one numerical variable only
- C. time alone
- D. class boundaries only

Correct answer: two categorical variables

Explanation and steps

Rows and columns represent categories.

Question 5 - MCQ · PG-49 · TH-Q5

Gender and food preference can be summarized using a

- A. contingency table
- B. frequency polygon only
- C. stem plot only
- D. single tally

Correct answer: contingency table

Explanation and steps

Both variables are categorical.

Question 6 - MCQ · PG-49 · TH-Q6

Any quantity calculated from data is called a

- A. statistic
- B. category
- C. class boundary
- D. tally mark

Correct answer: statistic

Explanation and steps

Mean is an example of a statistic.

Question 7 - MCQ · PG-50 · TH-Q7

Among 72 teenagers, 40 prefer chocolate. How many prefer vanilla?

- A. 32
- B. 18
- C. 40
- D. 58

Correct answer: 32

Explanation and steps

$$72 - 40 = 32.$$

Question 8 - MCQ · PG-50 · TH-Q8

If the sample has 180 persons and 72 teenagers, the number of adults is

- A. 108
- B. 72
- C. 90
- D. 180

Correct answer: 108

Explanation and steps

$$180 - 72 = 108.$$

Question 9 - MCQ · PG-50 · TH-Q9

If 58 adults prefer vanilla out of 108 adults, adults preferring chocolate number

- A. 50
- B. 32
- C. 40
- D. 90

Correct answer: 50

Explanation and steps

$$108 - 58 = 50.$$

Question 10 - MCQ · PG-50 · TH-Q10

Among 72 teenagers, 40 prefer chocolate and 32 prefer vanilla. Among 108 adults, 50 prefer chocolate and 58 prefer vanilla.

How many persons prefer chocolate?

- A. 90
- B. 72
- C. 108
- D. 180

Correct answer: 90

Explanation and steps

40 teenagers + 50 adults = 90.

Question 11 - MCQ · PG-50 · TH-Q11

Among 72 teenagers, 40 prefer chocolate and 32 prefer vanilla. Among 108 adults, 50 prefer chocolate and 58 prefer vanilla.

How many persons prefer vanilla?

- A. 90
- B. 72
- C. 108
- D. 180

Correct answer: 90

Explanation and steps

32 teenagers + 58 adults = 90.

Question 12 - MCQ · PG-51 · TH-Q12

The chi-square statistic is denoted by

- A. χ^2
- B. χ
- C. σ^2
- D. r

Correct answer: χ^2

Explanation and steps

Chi-square uses the Greek letter chi squared.

Question 13 - MCQ · PG-51 · TH-Q13

The chi-square statistic measures relationship between

- A. two categorical variables
- B. two means only
- C. one continuous variable only
- D. time and distance only

Correct answer: two categorical variables

Explanation and steps

It compares observed and expected categorical frequencies.

Question 14 - MCQ · PG-51 · TH-Q14

The chi-square statistic is always

- A. non-negative
- B. negative
- C. between -1 and 1
- D. equal to zero only

Correct answer: non-negative

Explanation and steps

Squared deviations divided by positive expected frequencies cannot be negative.

Question 15 - MCQ · PG-51 · TH-Q15

In $\chi^2 = \sum ((O_{ij} - E_{ij})^2 / E_{ij})$, O_{ij} denotes

- A. observed frequency
- B. expected frequency
- C. row total
- D. grand total

Correct answer: observed frequency

Explanation and steps

O stands for observed.

Question 16 - MCQ · PG-51 · TH-Q16

In the chi-square formula, E_{ij} denotes

- A. expected frequency
- B. observed frequency
- C. column heading
- D. sample mean

Correct answer: expected frequency

Explanation and steps

E stands for expected.

Question 17 - MCQ · PG-51 · TH-Q17

Expected frequency E_{ij} equals

- A. (row total $\times$ column total)/grand total
- B. observed frequency/grand total
- C. row total + column total
- D. grand total/number of rows

Correct answer: (row total $\times$ column total)/grand total

Explanation and steps

Use $E_{ij} = R_i C_j / N$.

Question 18 - MCQ · PG-51 · TH-Q18

In the expected-frequency formula, R_i denotes

- A. row total
- B. column total
- C. grand total
- D. observed cell

Correct answer: row total

Explanation and steps

R_i is the total of row i.

Question 19 - MCQ · PG-51 · TH-Q19

In the expected-frequency formula, C_j denotes

- A. column total
- B. row total
- C. grand total
- D. observed cell

Correct answer: column total

Explanation and steps

C_j is the total of column j .

Question 20 - MCQ · PG-51 · TH-Q20

In the expected-frequency formula, N denotes

- A. grand total
- B. number of rows only
- C. number of columns only
- D. one cell frequency

Correct answer: grand total

Explanation and steps

N counts all observations.

Question 21 - MCQ · PG-51 · TH-Q21

If observed and expected frequencies are identical in every cell, χ^2 equals

- A. 0
- B. 1
- C. -1
- D. N

Correct answer: 0

Explanation and steps

Every contribution is zero.

Question 22 - MCQ · PG-51 · TH-Q22

A larger difference between observed and expected frequencies generally gives

- A. a larger χ^2
- B. a negative χ^2
- C. $\chi^2 = 0$ always
- D. a smaller grand total necessarily

Correct answer: a larger χ^2

Explanation and steps

Squared deviations increase.

Question 23 - MCQ · PG-51 · TH-Q23

	Right-handed	Left-handed	Total
Men	40	40	80
Women	35	85	120
Total	75	125	200

The expected frequency for men who are right-handed is

- A. 30
- B. 40
- C. 45
- D. 50

Correct answer: 30

Explanation and steps

$$E_{11} = 80 \times 75 / 200 = 30.$$

Question 24 - MCQ · PG-51 · TH-Q24

	Right-handed	Left-handed	Total
Men	40	40	80
Women	35	85	120
Total	75	125	200

The expected frequency for men who are left-handed is

- A. 50
- B. 30
- C. 40
- D. 75

Correct answer: 50

Explanation and steps

$$E_{12} = 80 \times 125 / 200 = 50.$$

Question 25 - MCQ · PG-51 · TH-Q25

	Right-handed	Left-handed	Total
Men	40	40	80
Women	35	85	120
Total	75	125	200

The expected frequency for women who are right-handed is

- A. 45
- B. 35
- C. 50
- D. 75

Correct answer: 45

Explanation and steps

$$E_{21} = 120 \times 75 / 200 = 45.$$

Question 26 - MCQ · PG-51 · TH-Q26

	Right-handed	Left-handed	Total
Men	40	40	80
Women	35	85	120
Total	75	125	200

The expected frequency for women who are left-handed is

- A. 75
- B. 50
- C. 85
- D. 120

Correct answer: 75

Explanation and steps

$$E_{22} = 120 \times 125 / 200 = 75.$$

Question 27 - MCQ · PG-51 · TH-Q27

	Right-handed	Left-handed	Total
Men	40	40	80
Women	35	85	120
Total	75	125	200

The chi-square statistic is

- A. 8.88
- B. 6.55
- C. 3.37
- D. 0

Correct answer: 8.88

Explanation and steps

The four contributions sum to 8.88.

Question 28 - MCQ · PG-51 · TH-Q28

	Right-handed	Left-handed	Total
Men	40	40	80
Women	35	85	120
Total	75	125	200

The contribution from the cell $O = 40$, $E = 30$ is approximately

- A. 3.33
- B. 2.22
- C. 2.00
- D. 1.33

Correct answer: 3.33

Explanation and steps

$(40 - 30)^2/30 = 3.33$.

Question 29 - MCQ · PG-51 · TH-Q29

	Right-handed	Left-handed	Total
Men	40	40	80
Women	35	85	120
Total	75	125	200

The expected-frequency table preserves

- A. the same row and column totals
- B. only the grand total
- C. none of the totals
- D. only diagonal totals

Correct answer: the same row and column totals

Explanation and steps

Expected counts are constructed from the marginal totals.

Question 30 - MCQ · PG-52 · TH-Q30

	Pink	Blue	Orange	Total
Boys	27	63	10	100
Girls	41	45	14	100
Total	68	108	24	200

The expected frequency for boys choosing pink is

- A. 34
- B. 27
- C. 41
- D. 68

Correct answer: 34

Explanation and steps

$$100 \times 68 / 200 = 34.$$

Question 31 - MCQ · PG-52 · TH-Q31

	Pink	Blue	Orange	Total
Boys	27	63	10	100
Girls	41	45	14	100
Total	68	108	24	200

The expected frequency for boys choosing blue is

- A. 54
- B. 45
- C. 63
- D. 108

Correct answer: 54

Explanation and steps

$$100 \times 108 / 200 = 54.$$

Question 32 - MCQ · PG-52 · TH-Q32

	Pink	Blue	Orange	Total
Boys	27	63	10	100
Girls	41	45	14	100
Total	68	108	24	200

The expected frequency for boys choosing orange is

- A. 12
- B. 10
- C. 14
- D. 24

Correct answer: 12

Explanation and steps

$$100 \times 24 / 200 = 12.$$

Question 33 - MCQ · PG-52 · TH-Q33

	Pink	Blue	Orange	Total
Boys	27	63	10	100
Girls	41	45	14	100
Total	68	108	24	200

The expected frequencies for girls are

- A. 34, 54, 12
- B. 41, 45, 14
- C. 68, 108, 24
- D. 27, 63, 10

Correct answer: 34, 54, 12

Explanation and steps

Both gender row totals equal 100.

Question 34 - MCQ · PG-52 · TH-Q34

	Pink	Blue	Orange	Total
Boys	27	63	10	100
Girls	41	45	14	100
Total	68	108	24	200

The chi-square statistic is

- A. 6.55
- B. 8.88
- C. 3.37
- D. 0

Correct answer: 6.55

Explanation and steps

The six cell contributions sum to 6.55.

Question 35 - MCQ · PG-52 · TH-Q35

The symbol Σ in the chi-square formula means

- A. sum over all cells
- B. subtract all cells
- C. take only one cell
- D. find the mean

Correct answer: sum over all cells

Explanation and steps

Every cell contributes to χ^2 .

Question 36 - MCQ · PG-52 · TH-Q36

A 2×2 contingency table contains

- A. 4 data cells
- B. 2 data cells
- C. 6 data cells
- D. 8 data cells

Correct answer: 4 data cells

Explanation and steps

Two rows times two columns gives four.

Question 37 - MCQ · PG-52 · TH-Q37

A table with 2 row categories and 3 column categories contains

- A. 6 data cells
- B. 5 data cells
- C. 3 data cells
- D. 9 data cells

Correct answer: 6 data cells

Explanation and steps

$$2 \times 3 = 6.$$

Question 38 - MCQ · PG-52 · TH-Q38

Expected frequencies are calculated from

- A. marginal totals and the grand total
- B. cell labels only
- C. question number
- D. class width only

Correct answer: marginal totals and the grand total

Explanation and steps

They represent counts expected from the margins.

Question 39 - MCQ · PG-52 · TH-Q39

The chi-square calculation compares

- A. observed and expected frequencies
- B. means and medians
- C. class marks and widths
- D. maximum and minimum only

Correct answer: observed and expected frequencies

Explanation and steps

Each term uses $O - E$.

Question 40 - MCQ · PG-52 · TH-Q40

Contingency tables and χ^2 are especially suited to

- A. categorical data
- B. only continuous measurements
- C. only ordered raw scores
- D. only graphs

Correct answer: categorical data

Explanation and steps

Categories are counted into cells.